MILITARY SPECIFICATION SHEET

ELECTRON TUBES, CATHODE RAY

TYPES 5AQP1B, 5AQP2B, 5AQP7B, 5AQP11B, AND 5AQP31B

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Flat face, aluminized screen, electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: *See figure 1*

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Rg	Zd	Ehk	Alt
Unit:	V	Vdc	vdc	Vdc	Vdc	Meg	Meg	Vdc	ft
Maximum:	6.9	0	400	1,650	4,400	1.5	1.0	+180	40,000
Minimum:	5.7	-200	---	---	2,500	---	---	-180	---
TEST CONDITIONS:	6.3	Adj	---	Focus	3,500	---	---	---	---

GENERAL:

(A) Qualification - Not required

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
5201	Voltage breakdown	All		---	---	---	---	---	---
5201	Voltage breakdown (electrostatic types)	All		---	---	---	---	---	---
5206	Gas "cross"	All	See note 1						
		P1	Light = 25 ftL	---	---	---	---	---	---
		P2	Light = 23 ftL	---	---	---	---	---	---
		P7	Light = 8 ftL (see note 2)	---	---	---	---	---	---
		P11	Light = 20 ftL (see note 5)	---	---	---	---	---	---
		P31	Light = 40 ftL	---	---	---	---	---	---
5106	Bulb, screen, and faceplate blemishes	All		---	---	---	---	---	---
5231	Spot position (electrostatic deflection)	All		---	---	---	---	16	mm
5231	Spot displacement (leakage)	All		---	---	Displ	---	10	mm
5241	Grid-cutoff voltage	All		---	---	Eco	-47	-79	Vdc
5251	Grid No. 1 leakage current	All		---	---	---	---	---	---
5251	Anode No. 2 leakage current	All		---	---	---	---	---	---
5103	Pattern distortion	All	See note 3	---	---	---	---	---	---
---	Centering of scan	All	Focused trace, Ec1 at 10-volt drive, 1D2 3D4	---	---	---	±2 inches minimum useful scan from tube face center		---
	Quality conformance inspection, part 2								
1101	Secureness of base, cap, or insert	All		---	---	---	---	---	---
1301	Heater current	All		---	---	If	540	660	mA
5201	Electrode current (anode No. 1)	All	Ec1 = 0	---	---	Ib1	-15	15	μAdc
5201	Electrode current (cathode)	P1	Light = 25 ftL	---	---	Ik	---	150	μAdc
		P2	Light = 23 ftL	---	---	Ik	---	150	μAdc
		P7	Light = 8 ftL (see note 2)	---	---	Ik	---	150	μAdc
		P11	Light = 20 ftL (see note 5)	---	---	Ik	---	150	μAdc
		P31	Light = 40 ftL	---	---	Ik	---	150	μAdc
5101	Base alinement (electrostatic types)	All	+1D2, pin No. 5	---	---	---	---	---	---
5101	Neck and base alinement (electrostatic types)	All		---	---	---	---	---	---
5101	Angle between traces	All		---	---	---	89.2	90.8	Degrees

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
5216	Cathode illumination	All		---	---	---	---	---	---
5216	Stray light emission (conventional types)	All	Eb = 4,400 Vdc	---	---	---	---	---	---
(A) 5221	Screen	P7		---	---	D1(5) G5:1	400 4.0	--- ---	cb ---
5223	Modulation	P1	Light = 25 ftL	---	---	ΔEc1	---	25	Vdc
		P2	Light = 23 ftL	---	---	ΔEc1	---	25	Vdc
		P7	Light = 8 ftL (see note 2)	---	---	ΔEc1	---	25	Vdc
		P11	Light = 20 ftL (see note 5)	---	---	ΔEc1	---	25	Vdc
		P31	Light = 40 ftL	---	---	ΔEc1	---	25	Vdc
5226	Line width A (electrostatic deflection)	P1	Light = 25 ftL	---	---	Width	---	0.35	mm
		P2	Light = 23 ftL	---	---	Width	---	.35	mm
		P7	Light = 8 ftL (see note 2)	---	---	Width	---	.35	mm
		P11	Light = 20 ftL (see note 5)	---	---	Width	---	.35	mm
		P31	Light = 40 ftL	---	---	Width	---	.35	mm
5226	Line width B (electrostatic deflection)	All	See note 4						
		P1	Light = 25 ftL	---	---	Width	---	0.42	mm
		P2	Light = 23 ftL	---	---	Width	---	.42	mm
		P7	Light = 8 ftL (see note 2)	---	---	Width	---	.42	mm
		P11	Light = 20 ftL (see note 5)	---	---	Width	---	.42	mm
		P31	Light = 40 ftL	---	---	Width	---	.42	mm
5246	Focusing voltage at cutoff	All		---	---	Eb1	---	420	Vdc
5246	Focusing voltages, zero bias	All		---	---	Eb1	0	---	Vdc
5248	Deflection factor	All	1D2	---	---	DF	57	69	Vdc/in.
5248	Deflection factor	All	3D4	---	---	DF	44	54	Vdc/in.
5251	Heater-cathode leakage current	All		---	---	---	---	10	μAdc
(A) 1105	Permanence of marking	All		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life-test provisions	All	Group C; t = 500 hrs (min); Eb2 = 4,400 Vdc						
		P1	Light = 25 ftL	---	---	---	---	---	---
		P2	Light = 23 ftL	---	---	---	---	---	---
		P7	Light = 8 ftL (see note 2)	---	---	---	---	---	---
		P11	Light = 20 ftL (see note 5)	---	---	---	---	---	---
		P31	Light = 40 ftL	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3 -Continued								
---	Life-test end points	All	See note 4						
		P1	Light = 22 ftL	---	---	---	---	---	---
		P2	Light = 20 ftL	---	---	---	---	---	---
		P7	Light = 6 ftL (see note 2)	---	---	---	---	---	---
		P11	Light = 18 ftL (see note 5)	---	---	---	---	---	---
		P31	Light = 35 ftL	---	---	---	---	---	---
			Line width A	---	---	Width	---	0.35	mm
			Line width B	---	---	Width	---	0.42	mm
			Modulation	---	---	Ec1	---	25	Vdc

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. Measured in accordance with method 5221, using a Wratten No. 15 filter, or equivalent.
3. All portions of a raster pattern, adjusted so its widest points just touch the sides of a 4-inch (101.6 mm) square, will fall within the area bounded by the 4-inch (101.6 mm) square and an inscribed 3.916-inch (99.5 mm) square, except when geometry of the tube makes this impossible.
4. Measured at a distance from the center of the screen equal to 1/3 of the maximum bulb diameter.
(A) 5. Measured in accordance with method 5221 using a type 3 photronic cell without eye correction, calibrated in footcandles of illumination from a light source having a color temperature of 2, 700°K.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Navy -
Air Force - 11, 17
DSA - ES

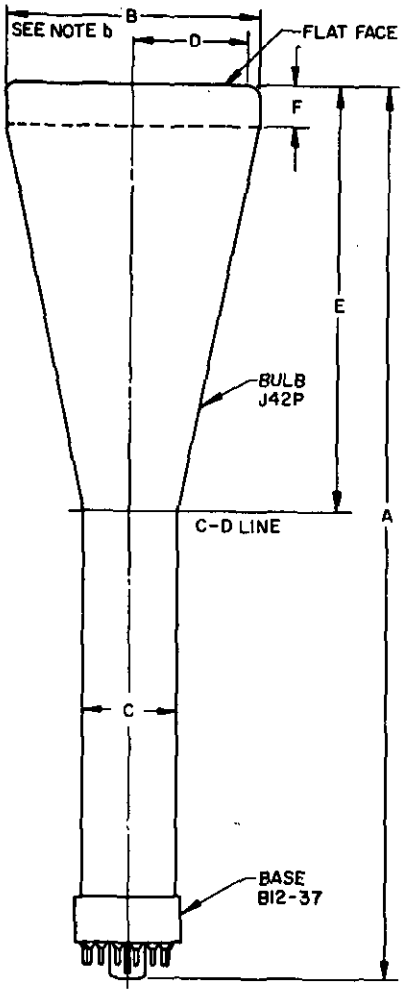
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

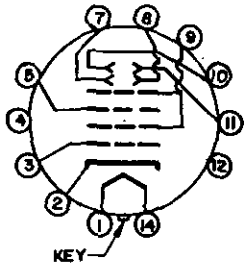
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PIN CONNECTIONS

Pin No.	Element
1	Heater
2	Cathode
3	Grid No. 1
4	Internal connection
5	Focusing electrode
7	Deflecting electrode D_3
8	Deflecting electrode D_4
9	Accelerator
10	Deflecting electrode D_2
11	Deflecting electrode D_1
12	NC
14	Heater



BASING DIAGRAM

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	*Maximum*
Quality conformance inspection, part 2		
A	16.563 (420.70)	16.938 (440.23)
B	5.156 (130.96)dia.	5.344 (135.74)dia.
C	1.938 (49.23)dia.	2.063 (52.40)dia.
D	2.250 (57.15)R	
Reference dimensions (see note a)		
E	9.000 (228.60)	
F	0.875 (22.23)	

NOTES:
a. These dimensions are for information only and are not required for inspection purposes.
b. Useful screen.

FIGURE 1. Outline drawing of electron tube types 5AQP1B, 5AQP2B, 5AQP7B, 5AQP11B, and 5AQP31B.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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